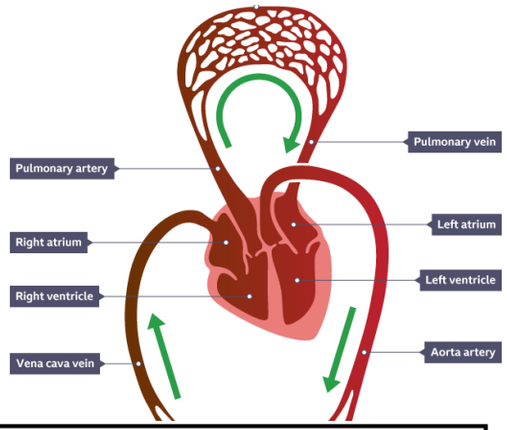
Knowledge Organiser
Year 8: Human Body – The Circulatory System

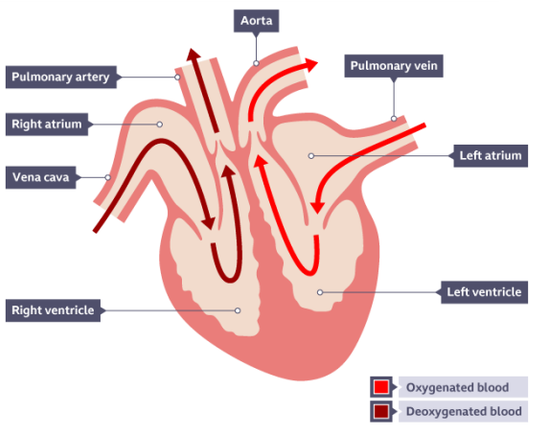
What is the circulatory system?
- The circulatory system is the **heart** and all the **blood vessels** in the body which carry cells and substances to where they are needed
- There are three types of blood vessel: arteries, veins and capillaries
- **Arteries** carry blood away from the heart under high pressure
- **Veins** carry blood back to the heart
- **Capillaries** are tiny vessels that branch out into tissues carrying substances like oxygen

A diagram showing the basic structure of the human circulatory system



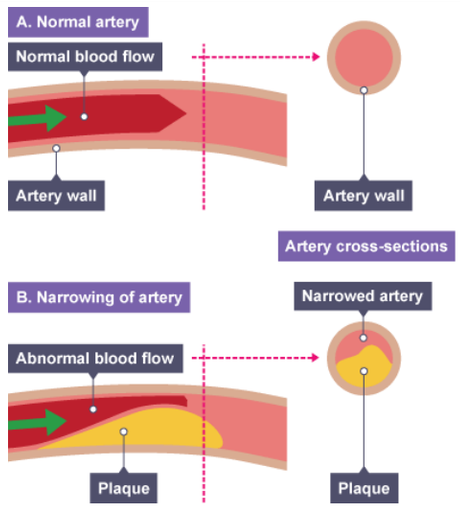
How does the heart work?
- The heart is an **organ** mostly made of **muscle tissue** which pumps blood around the body
- The heart has **four chambers** and two non-symmetrical sides: right and left
- Electrical signals trigger the different parts of the heart to **contract in sequence** to create a pressure wave

A labelled diagram of the heart shown looking outwards from the chest



The stages of a heart beat…
1. Deoxygenated blood enters the heart in the right atrium
2. The right atrium contracts, pushing the blood into the right ventricle
3. The right ventricle contracts sending the blood into the lungs
4. Oxygen diffuses into the blood in the lungs and re-enters the heart in the left atrium
5. The left atrium contracts pushing the blood into the left ventricle
6. The left ventricle contracts and the high pressure oxygenated blood is pushed around the body

How does diet affect heart function?
- An unhealthy diet will affect the function of the heart
- A diet high **in saturated fats** can lead to fatty deposits called plaque building up in the artery
- The coronary artery supplies **oxygenated blood** to the muscles in the heart itself
- If this gets blocked and oxygen cannot get to the muscle tissue, the heart cannot continue to contract, this causes a **heart attack**



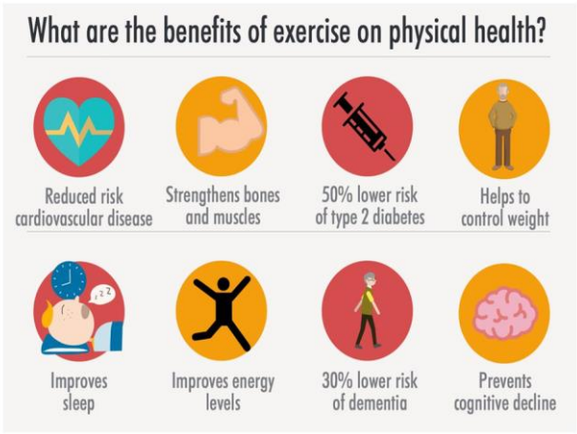
A diagram showing the narrowing of an artery caused by fat deposits

The effect of stimulants on heart function:
- Stimulants are a class of drugs that raise levels of activity in the central nervous system
- They include; caffeine, cocaine and nicotine
- Stimulants temporarily make the heat beat at a faster rate
- The heart also beats with a greater force, raising blood pressure throughout the body
- This can results in an increased risk of heart attack and stroke for already unhealthy people



Why does exercise effect heart rate?
- During exercise the **heart beats faster**
- Cells around the body need more energy during exercise, which is **released through respiration**
- This means that blood needs to be transferred around the body more quickly
- So the heart has to beat faster, sending more oxygen rich blood to the cells for **respiration**

A chart showing the effects of regular exercise on physical health:



RAG Yourself!

Learning Outcomes;	RAG
Describe the function of arteries and veins	
Draw a labelled diagram of the heart	
Explain why a high fat diet leads to high blood pressure	
Describe the stages of the function of the heart	
Explain the impact of exercise on the heart	

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A diagram showing the basic structure of the human circulatory system

How does the heart work?

A labelled diagram of the heart shown looking outwards from the chest

The stages of a heart beat…

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The effect of stimulants on heart function:



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